NAME:

a	b	c	d	e	Σ

Test 1: Propositional logic

Time limit: 45 minutes. Total points: 100.

Al Gore, George W. Bush, and Bill Clinton were asked about their stance on a new proposal in the area of fishing regulation. Their stance is influenced by the following relationships:

(i) If Al Gore supports a proposal, then Bush or Clinton (or both) must also support it.

(ii) Clinton will never support a proposal unless Al Gore also supports it.

(iii) Bush and Clinton cannot both support the same proposal.

Furthermore, we know that:

(iv) Every proposal is supported by at least one of these three politicians.

We want to show that:

(v) Bush supports this proposal if and only if Clinton does not support it.

(a) Formalize these statements as propositions $\varphi_1, \ldots, \varphi_5$ in the language $\mathbb{P} = \{a, b, c\}$, where a, b, c represent *"Al Gore/Bush/Clinton supports the proposal"*. *(10 pts)*

Let us further denote $T = \{\varphi_1, \varphi_2, \varphi_3, \varphi_4\}$.

(b) Using the resolution method, prove that $T \models \varphi_5$. (Write the corresponding set of clauses S in set representation, draw a resolution tree corresponding to some resolution refutation of S.) *(30 pts)*

(c) Using the tableau method, prove that $T \models \varphi_5$. (Construct a tableau proof from T, according to the definition from the lecture.) *(30 pts)*

(d) Consider the theory $T' = \{a \vee b\}$ in the language $\mathbb{P}' = \{a, b\}$. Is the theory T an extension of the theory T'? If so, is it a conservative extension? Justify your answers. *(15 pts)*

(e) Determine the number (up to equivalence) of propositions in the language $\mathbb{P}$ that are valid in the theory T, but not valid in the theory T'. Justify. Give one such proposition, if it exists. *(15 pts)*